

BIO-302 FINAL TERM- 100%

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MOLECULAR BIOLOGY

BY SULMAN ALI

0341-7547044

SUBJECTIVES

Functions of A site p site and E site.

1= A site: Aminoacyl-tRNA binding site in ribosome, where new amino acids are added to the growing polypeptide chain. **2= P site:** Peptidyl-tRNA binding site in ribosome, where the growing polypeptide chain is transferred to the incoming amino acid. **3= E site:** Exit site in ribosome, where the empty tRNA exits after releasing its amino acid.

Classes of splicing enlist:

There three classes of splicing including:

1= Nuclear pre-mRNA splicing: In eukaryotic cells, removal of non-coding introns and joining of exons to form mature mRNA.

2= Group II introns splicing: Self-splicing RNA elements found in certain organellar genomes that catalyze their own removal from precursor RNA.

3= Group I introns splicing: Self-splicing RNA sequences, often in tRNA or rRNA genes, that excise themselves from precursor RNA in a ribozyme-mediated process.

What is the Plasmid and where it is present.

A plasmid is a small, circular DNA molecule separate from the chromosomal DNA, found in the cytoplasm of the bacteria and some other organisms. It often carries accessory genetic information and can replicate independently, playing a role in gene transfer, antibiotic resistance, and genetic engineering.

Name two proteins necessary for lactose metabolism.

1= β -galactosidase: Converts lactose into glucose and galactose during lactose metabolism.

2= Lactose permease: Facilitates the entry of lactose into the bacterial cell for metabolism.

Initiation of transcription in prokaryotes:

Transcription initiation in prokaryotes involves RNA polymerase recognizing the promoter region on DNA, where it binds and unwinds the double helix. The sigma factor of RNA polymerase helps in promoter recognition. This forms a transcription bubble. RNA polymerase then begins synthesizing RNA complementary to the DNA template strand. As the transcription complex moves along the DNA, the DNA helix re-forms and the RNA molecule is elongated, initiating the synthesis of the mRNA transcript.

Group I intron splicing:

Group I intron splicing involves a self-excising RNA sequence found in tRNA or rRNA genes. The intron folds into a complex structure with catalytic properties, forming a ribozyme. It undergoes conformational changes, requiring an external guanosine nucleotide as a cofactor. The 3'-OH of this guanosine acts as a nucleophile, cleaving the intron from the precursor RNA. This links the splice donor site to the splice acceptor site, releasing the intron and joining the exons. This process exemplifies RNA's enzymatic capability.

How and why methylation of N-3 in adenine is significant in treatment of cancer.

Methylation at N-3 in adenine is of great practical significance because N-3 methyl adenine formation blocks DNA replication but does not appear to lead to mutations. Therefore, a methylating agent that can transfer methyl group exclusively to N-3 position in adenine would have the potential to kill cancer cells without causing mutations.

What is purpose of transient excursion in transcription.

The "transient excursion" model proposes transient cycles of forward and reverse translocation of RNA polymerase. Thus, polymerase is thought to leave the promoter and translocate a short way along the DNA template, synthesizing a short transcript before aborting transcription, releasing the transcript, and returning to its original location on promoter.

What is alternative splicing and its importance.

Alternative splicing is a process in which a single pre-mRNA can be spliced into different mature mRNA molecules by including or excluding certain exons. This diversifies the protein products from a single gene, enhancing genetic complexity, functional diversity, and adaptability in organisms, playing a critical role in cellular specialization and disease development.

How transition to open complex are worked.

Transition to open complex in transcription involves RNA polymerase binding to the promoter region and undergoing conformational changes. Initially, the closed complex forms between polymerase, DNA, and a sigma factor. Then, local DNA unwinding occurs near the transcription start site, creating the open complex. This involves DNA melting, with the template strand separated to allow RNA synthesis. The polymerase transitions from initiation to elongation mode, beginning mRNA synthesis.

What is ADAR, name 3 genes that codes ADAR in humans:

ADAR (Adenosine Deaminase Acting on RNA) is an enzyme that catalyzes the conversion of adenosine to inosine in RNA molecules, affecting gene expression and function. In humans, ADAR is coded by three genes: ADAR1, ADAR2, and ADAR3. These enzymes play roles in RNA editing, impacting protein diversity, nervous system function, and disease development, such as in neurodevelopmental disorders and cancer.

What is nitrogenous bases are included in purine and pyrimidine.

Purines and pyrimidines are nitrogenous bases found in DNA and RNA. Purines include adenine (A) and guanine (G), featuring a double-ring structure. Pyrimidines consist of cytosine (C), thymine (T, found in DNA), and uracil (U, found in RNA), with a single-ring structure. These bases pair in specific combinations (A-T and G-C in DNA, A-U and G-C in RNA) to form the genetic code.

Define Topoisomerase:

Topoisomerase is an enzyme that alters the topology of DNA by adding or removing twists and knots, facilitating DNA replication, transcription, and repair while preventing excessive supercoiling.

Define the Elongation step of transcription:

Elongation is the second phase of transcription where RNA polymerase moves along the DNA template strand, synthesizing complementary RNA, and reading the DNA code to create a growing mRNA transcript.

Chargaff's rule:

Chargaff's rule, formulated by biochemist Erwin Chargaff, states that in DNA, the amounts of adenine (A) are equal to thymine (T), and the amounts of guanine (G) are equal to cytosine (C). This base pairing specificity is fundamental to the double-stranded nature of DNA, as adenine pairs with thymine via two hydrogen bonds, and guanine pairs with cytosine via three hydrogen bonds, ensuring structural stability and accurate replication of genetic information.

How many CTD heptatide in yeast human and drosophila.

There are 27 of these repeats in the yeast Pol II CTD, 32 in the worm *Caenorhabditis elegans*, 45 in the fly *Drosophila*, and 52 in humans.

DNA helicase:

DNA helicase is an enzyme crucial in DNA replication and repair processes. It unwinds the double-stranded DNA molecule by breaking the hydrogen bonds between the complementary base pairs, separating the two strands and creating a replication fork. This unwinding allows other enzymes and proteins, such as DNA polymerase, to access the single strands and synthesize new DNA strands during replication or repair events.

Describe the Meselson and Stahl full experiment:

The Meselson and Stahl experiment provided evidence for the semi-conservative replication of DNA. They cultured *E. coli* in a medium containing "heavy" nitrogen (^{15}N) to label the DNA. After allowing replication in a "light" nitrogen (^{14}N) medium, DNA samples were taken and subjected to density gradient centrifugation. The results showed intermediate-density DNA after one generation and eventually a complete shift to light DNA, supporting the semi-conservative replication model proposed by Watson and Crick.

Transition factors:

Transition factors are proteins that assist RNA polymerase in transitioning from initiation to elongation during transcription. They help polymerase escape the promoter region and begin synthesizing the RNA transcript.

Mismatched repair system:

Mismatch repair is a cellular process that corrects errors arising during DNA replication, rectifying mismatches and small insertion/deletion loops. Enzymes recognize the mismatch, remove the incorrect nucleotide, and resynthesize the correct DNA sequence, maintaining genomic integrity.

Characteristics of genetic material:

Genetic material possesses traits like heredity, encoding information in nucleotide sequences, replication to pass information to offspring, mutability enabling evolution, and the ability to express information as proteins, determining an organism's traits and functions.

How insertion and deletion change DNA sequence.

Insertion adds extra nucleotides to the DNA sequence, shifting the reading frame and potentially altering all downstream codons. Deletion removes nucleotides, also causing a frame shift and impacting subsequent codons. Both can lead to non-functional or altered proteins.

How molecular biology has shown advancements in agriculture:

Molecular biology has advanced agriculture through genetic engineering, allowing the modification of crop traits for increased yield, pest resistance, and nutritional content, addressing global food challenges sustainably.

why plasmids are not regarded as cell's genome:

They are not regarded as part of the cell's genome for two reasons.

1= First, a particular plasmid may be found in cells of different species and may move from one host species to another.

2= Second, a plasmid may sometimes be present and sometimes absent from the cells of a particular host species. They are not needed for cell growth and division under normal conditions.

Why mRNA has a short life span:

mRNA has a short lifespan due to enzymatic degradation by ribonucleases. This rapid turnover allows cells to regulate gene expression swiftly, adapting to changing environmental conditions and preventing the accumulation of obsolete genetic information.

Aflatoxin B1:

Aflatoxin B1, the most potent toxin produced by *A. flavus* presents a particularly serious health threat in the United States. Aflatoxin B1 can induce DNA damage through its metabolites, forming adducts with DNA bases, primarily guanine. These adducts interfere with DNA replication and repair mechanisms, leading to mutations and disruptions in gene expression. This DNA damage can contribute to the development of cancers and other health problems, emphasizing the significance of controlling aflatoxin exposure.

RNA processing event:

RNA processing involves modifications to primary transcripts. In eukaryotes, pre-mRNA undergoes splicing to remove introns and join exons. A 5' cap is added for stability, and a poly-A tail at the 3' end. These modifications ensure mature mRNA stability, transport, and accurate translation, influencing gene expression and protein synthesis.

Five major features of the operon model:

An operon is a functional unit of bacterial DNA containing a group of genes under the control of a single promoter.

1. The products of the *lacZ*, *lacY* and *lacA* genes are encoded in a single polycistronic *lac* mRNA molecule.
2. The promoter for this mRNA molecule is immediately adjacent to the *lac* region. Promoter mutations (*p-*) that are completely incapable of making β galactosidase, permease, and transacetylase have been isolated. The promoter is located between *lacI* and *lacO*.
3. The operator is a sequence of bases (in the DNA) to which the repressor protein binds.
4. When the repressor protein is bound to the operator, *lac* mRNA transcription can't take place.
5. Inducers stimulate *lac* mRNA synthesis by binding to the repressor. This binding alters the repressor's conformation so it can't bind to the operator. In the presence of an inducer, therefore, The operator is unoccupied and the promoter is available for initiation of mRNA synthesis. This state is called derepression.

Mutations can change base one to another ? Name any two Mutations name

Mutations alter DNA sequences by changing one base to another. Two examples are:

Missense Mutation: A single base change results in a different amino acid in the protein.

Nonsense Mutation: A base change creates a premature stop codon, leading to a truncated and often nonfunctional protein.

Initiation in prokaryotes:

In prokaryotes, initiation refers to the beginning stage of transcription. It involves the binding of RNA polymerase to the promoter region of DNA, forming a closed complex. This is followed by the transition to an open complex, allowing for RNA synthesis to begin.

Damage of alkylation:

Alkylation damage is a type of DNA damage caused by the addition of alkyl groups to DNA bases, altering their structure and disrupting normal base pairing. This can lead to errors during replication and transcription, potentially resulting in mutations and genomic instability.

Define promoter, oligomer, and multimer:

A promoter is a DNA sequence initiating gene expression. An oligomer is a short chain of repeating molecules, like nucleotides. A multimer refers to a complex consisting of multiple identical or similar subunits.

What is importance of nucleosome in transcription machinery?

Nucleosomes aid transcription by regulating DNA accessibility. They can be repositioned, enabling transcription factors and RNA polymerase to interact with DNA, facilitating gene expression while maintaining chromatin structure and integrity.

End replication problem:

The requirement for an RNA primer to initiate all new DNA synthesis creates a dilemma for the replication of the ends of linear chromosomes. This is called as the end replication problem.

Name of the experiment in Messenger RNA processing ?

The experiment related to Messenger RNA (mRNA) processing is called "RNA splicing." This process involves removing introns (non-coding regions) and joining exons (coding regions) to generate a mature and functional mRNA molecule.

Three significant discoveries of genetics:

1. Mendel's laws (1865) elucidated hereditary traits transmission.
2. Watson and Crick's DNA structure (1953) revealed genetic information storage.
3. Human Genome Project (2003) sequenced entire DNA, advancing personalized medicine and genetic research.

Errors in DNA replication:

DNA replication errors, or mutations, result from mistakes in copying DNA during cell division. Substitutions, deletions, or insertions of nucleotides can alter the genetic code, potentially leading to dysfunctional proteins, diseases, or evolutionary changes.

What are the types of splicing:

1. **Constitutive Splicing:** Standard process where exons are joined and introns are removed.
2. **Alternative Splicing:** Exons can be selectively included/excluded, generating diverse mRNA isoforms, expanding proteomic complexity and functional versatility.

How the change in nucleosome effects transcription factors.

Changes in nucleosome positioning or modifications can alter DNA accessibility. This impacts transcription factors' ability to bind specific gene regions, either enhancing or inhibiting gene expression. Remodeled nucleosomes facilitate or hinder transcription factor interactions, influencing gene regulation.

How nucleosome structure change help transcription machinery:

Nucleosome structure changes, like sliding or eviction, expose promoter regions, allowing transcription machinery access to DNA. This enhances gene transcription by enabling

transcription factors and RNA polymerase to bind and initiate the transcription process effectively.

what is the structure of CPD and also write its function?

The structure of a Cyclobutane Pyrimidine Dimer (CPD) involves the covalent bonding of two adjacent pyrimidine bases (thymine or cytosine) within DNA, forming a cyclobutane ring. CPDs are formed as a result of UV radiation exposure. Their function is to induce DNA repair mechanisms, such as CPD Photolyase, to remove and correct these UV-induced DNA lesions and maintain genomic integrity.

What is the role of mRNA in a cell?

mRNA carries genetic information from DNA to ribosomes, guiding protein synthesis. It serves as a transient intermediary, ensuring accurate translation of DNA's instructions into functional proteins that drive cellular processes and functions.

Explain semi conservative model of DNA replication?

The semi-conservative model of DNA replication proposes that during cell division, each parental DNA strand serves as a template for the synthesis of a new complementary strand. This results in two daughter DNA molecules, each conserving one original and one newly synthesized strand.

What are the Primer and its functions.

A primer is a short RNA or DNA sequence that initiates DNA synthesis during replication or PCR. It provides a starting point for DNA polymerase to add complementary nucleotides, enabling the synthesis of a new DNA strand.